

MOLECULAR BIOLOGY PROJECT · CANCER GENOMICS

Decoding Breast Cancer

Subtypes, Mutations & the Path to New Therapies

An investigation using public cancer databases

cBioPortal · **Enrichr** · **STRING** · **Excel / Google Sheets**

An introductory briefing





PROJECT OVERVIEW

What You Will Do in This Project

You will act as a cancer genomics researcher: pull real patient data from public databases, compare breast cancer subtypes at the molecular level, and interpret what your findings could mean for treatment.

1

Ask

Frame a research question comparing ER+, HER2+, and triple-negative tumors



2

Extract

Pull mutation & expression data for chosen genes from cBioPortal



3

Analyze

Summarize patterns and build visualizations in Excel / Sheets



4

Interpret

Use Enrichr & STRING to find affected pathways and protein networks



5

Present

Deliver a scientific poster or presentation of your findings



Cancer Is a Disease of the Genome

Cancer begins when a normal cell accumulates changes in its DNA that let it ignore the body's rules for growth.

It divides when it shouldn't, resists signals to stop or die, and can invade nearby tissue or spread (metastasize).

Because the root cause is molecular, we can read a tumor's DNA to understand — and eventually target — what went wrong.

Why data matters

Every tumor carries a unique combination of DNA changes. Large public databases have sequenced thousands of them — turning individual biology into patterns we can study.

SELECT HALLMARKS OF CANCER



Sustained growth signals



Evading growth suppressors



Unlimited replication



Inducing blood supply



Invasion & metastasis



Resisting cell death



The Two Kinds of Cancer Genes

Not every mutation matters. A few driver mutations push a cell toward cancer; most are passenger mutations along for the ride. Drivers usually fall into two opposite categories:



Oncogenes

"Stuck accelerator"

- Normal role: signal cells to grow and divide
- In cancer: a gain-of-function mutation locks them ON
- One activated copy is enough to drive growth
- Breast cancer examples: ERBB2 (HER2), PIK3CA, MYC



Tumor Suppressors

"Failed brakes"

- Normal role: restrain growth, repair DNA, trigger cell death
- In cancer: a loss-of-function mutation switches them OFF
- Usually, both copies must be lost to remove the brake
- Breast cancer examples: TP53, BRCA1/2, PTEN, RB1



In cBioPortal you will see these directly: recurrent mutations across many patients often mark a driver gene worth investigating as a drug target.



WHY BREAST CANCER

A Common Disease That Is Not One Disease

Breast cancer is among the most common cancers worldwide — and one of the best studied. Decades of research show it is really a family of biologically distinct diseases that look similar under the microscope but behave very differently.

~1 in 8

U.S. women develop breast cancer in their lifetime

3

major clinically defined subtypes you will compare

1000s

tumors already sequenced and free to explore online

~5

genes are enough for a focused analysis



The Three Subtypes You Will Compare

Clinicians classify breast tumors by three receptors on the cell surface. Which receptors are present determines both biology and treatment — and defines the three groups at the heart of your project.

ER-Positive

Luminal (A / B) · Most common (~70%)

- Fueled by estrogen (ER+ / often PR+)
- Generally slower-growing, better outlook
- Frequent PIK3CA mutations; GATA3
- Treated with hormone (endocrine) therapy

HER2-Positive

HER2-enriched · ~15-20%

- Extra copies of the HER2 (ERBB2) gene
- Historically aggressive without treatment
- HER2 amplification is the key driver
- Transformed by HER2-targeted antibodies

Triple-Negative

Often basal-like · ~10-15%

- ER-, PR-, and HER2- (no receptors)
- Most aggressive; younger patients, BRCA link
- Very high TP53 mutation rate
- Chemo, PARP inhibitors, immunotherapy



Beneath the Clinic: Molecular Subtypes

Gene-expression profiling (the PAM50 / intrinsic classification) sorts tumors by which genes are switched on. These molecular subtypes overlap with, but refine, the clinical labels and are exactly the kind of pattern you can explore in expression data.

Molecular subtype	Defining signature	Outlook
● Luminal A	ER+ / HER2-, low proliferation	Best prognosis
● Luminal B	ER+, higher proliferation	Intermediate
● HER2-enriched	HER2 high, often ER-	Targetable driver
● Basal-like	Overlaps triple-negative	Most aggressive



Clinical vs. molecular

The two systems mostly agree but not perfectly. A tumor called triple-negative in the clinic is usually, though not always, basal-like by expression.

Takeaway: molecular data can reveal biology that receptor status alone hides, and a core reason this project exists.



CELL OF ORIGIN

Where the Subtype Names Come From

These names are not random labels — they describe **which normal breast cell a tumor most resembles** in its gene-expression pattern. A breast duct is built from two cell layers, and that anatomy is the whole idea.

A BREAST DUCT, IN CROSS-SECTION



-  **Basal cells** — outer layer; structural, receptor-negative
-  **Luminal cells** — inner lining; sense hormones (ER / PR)
-  **Lumen** — the hollow channel at the center



Luminal A

Resembles the inner luminal cells. ER+ / PR+, HER2-, low proliferation. The slowest-growing subtype with the best outlook — and the most common.



Luminal B

Also luminal and ER+, but dividing faster (higher Ki-67). More aggressive than Luminal A, and often needs chemotherapy alongside hormone therapy.



Basal-like

Resembles the outer basal cells. Usually ER- / PR- / HER2-, so it overlaps heavily with triple-negative. High-grade, fast-growing, with TP53 and BRCA1 links.



Inflammatory Breast Cancer: A Different Axis

You may see "inflammatory" breast cancer mentioned as if it were a fifth molecular subtype. **It isn't.** IBC is a clinical diagnosis based on how a tumor presents and behaves — it can occur within Luminal A, Luminal B, HER2-enriched, or Basal-like disease.

DEFINED BY PRESENTATION, NOT EXPRESSION

Rapid onset — redness and swelling develop over weeks, often with no distinct lump

Skin changes — diffuse erythema and pitting ("peau d'orange") across a third or more of the breast

Underlying cause — tumor cells physically block the skin's dermal lymphatic vessels

Confirmed by biopsy — pathologists look for dermal lymphatic tumor emboli, not a gene-expression signature

Why this matters for your project: IBC is rare (roughly 1-5% of breast cancers) but aggressive. In cBioPortal, look for it as a clinical/histologic label — not a molecular-subtype column.

IT CUTS ACROSS EVERY MOLECULAR SUBTYPE



● Luminal A — 32%

● Luminal B — 18%

● HER2-enriched — 24%

● Basal-like — 24%

● Normal-like — 2%

Source: prospective INFLAME registry (RNA-seq, PAM50-classified IBC cases), 2025.

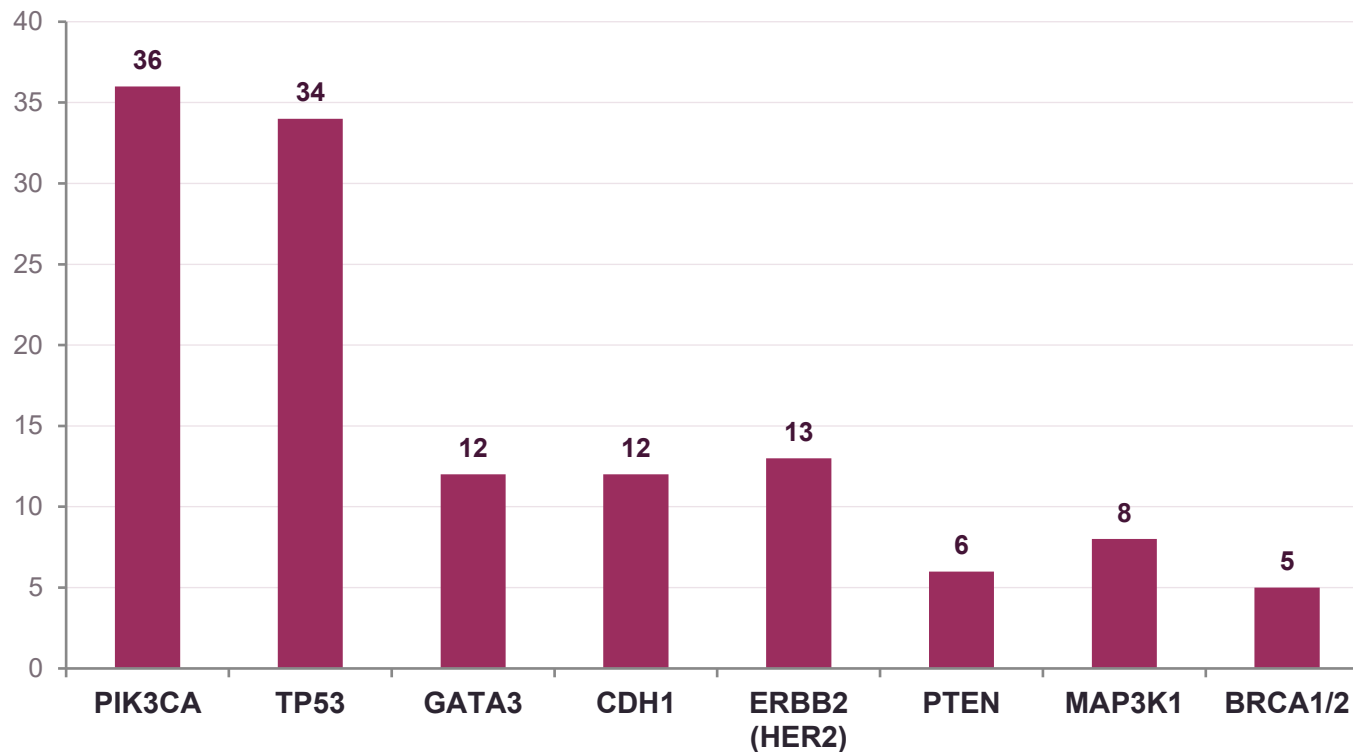
Older studies emphasized a skew toward basal-like and HER2-enriched disease; **this more recent registry shows IBC appearing across all four subtypes** — a reminder that these numbers evolve as more data is collected.



MUTATIONS BY GENE

The Usual Suspects: Key Breast Cancer Genes

A handful of genes are altered repeatedly in breast tumors. These make excellent starting points for your analysis. Approximate alteration frequencies across breast cancer are shown below.



READ IT BY SUBTYPE

PIK3CA & GATA3 — enriched in ER-positive tumors

TP53 — dominates triple-negative / basal-like disease

ERBB2 (HER2) — amplification defines the HER2+ group

BRCA1/2 — germline loss links to triple-negative & inherited risk



COMPARISON

Side by Side: What Sets the Subtypes Apart

Feature	ER-Positive	HER2-Positive	Triple-Negative
Key driver	ER signaling, PIK3CA	HER2 (ERBB2) amplification	TP53 loss, genomic instability
Typical growth	Slower	Fast (untreated)	Fast, aggressive
Main therapy	Endocrine + CDK4/6 inhibitors	Anti-HER2 antibodies	Chemo, PARP, immunotherapy
Relative outlook	Most favorable	Good with targeted therapy	Most challenging
Good genes to probe	ESR1, PIK3CA, GATA3	ERBB2, GRB7, TOP2A	TP53, BRCA1/2, PTEN



Where the Data Comes From

You don't need a lab to do this research — the data already exist. Major consortia have sequenced thousands of tumors and released the results for anyone to explore.



TCGA

The Cancer Genome Atlas

A landmark U.S. project that mapped DNA mutations, copy-number changes, and gene expression across 30+ cancer types — including a large breast cohort (BRCA).



METABRIC

Molecular Taxonomy of Breast Cancer

~2,500 breast tumors with expression and clinical data, widely used to study subtypes and long-term outcomes.



cBioPortal

Your window into both

A free, browser-based portal that lets you query these datasets by gene — no coding or downloads required to get started.



YOUR MAIN TOOL

Working in cBioPortal, Step by Step

1

Pick a study

Select a breast cancer dataset (e.g. Breast Invasive Carcinoma, TCGA or METABRIC) from the study list.

2

Enter your genes

Type in the genes you're investigating — for example TP53, PIK3CA, ERBB2 — as your query.

3

Read the OncoPrint

See a visual grid of which patients carry mutations, amplifications, or deletions in each gene.

4

Check frequencies

Use the Mutations and Cancer Types Summary tabs to read how often each gene is altered.

5

Export the data

Download results as a table and open it in Excel or Google Sheets for your own analysis.



What to record

- Gene name and its alteration frequency
- Which subtype or study the number came from
- Type of alteration (mutation, amplification, deletion)
- The date and dataset version you accessed
- A screenshot of each key chart

Good record-keeping is what makes your results reproducible.



INTERPRETATION TOOLS

From a Gene List to Biological Meaning

A list of mutated genes is a start, not an answer. The real question is: what do these genes do together? Two free tools turn your gene list into biology.



Enrichr

Pathway & enrichment analysis

- Paste in your gene list
- It tests whether those genes cluster in known pathways or functions
- Answers: "Are my genes involved in cell-cycle control? DNA repair? Hormone signaling?"
- Highlights biology that is over-represented in your set



STRING

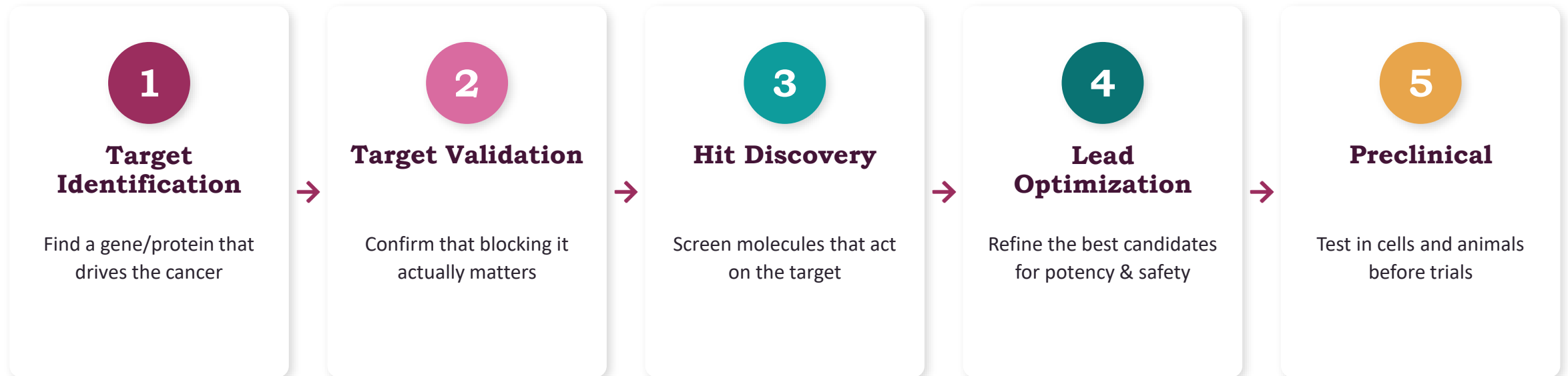
Protein-protein interaction networks

- Enter your genes/proteins
- It draws a network of which proteins physically or functionally interact
- Tightly connected hubs often point to a shared process — or a promising target
- Turns a flat list into a visual map of relationships



How Discoveries Become Drugs

Understanding tumor biology is the first link in a long chain. Drug discovery moves from biology to a candidate molecule long before any clinical trial. Your project lives at the very front of this pipeline.



Your project = Steps 1-2. You identify and begin to validate targets.



Reality check: bringing one drug to market can take 10+ years and cost billions — which is why smart early targeting matters.



Why Your Analysis Is Real Target Finding

A good drug target is a molecule that (1) drives the cancer and (2) can be safely blocked. The exact data you'll gather is what researchers use to nominate targets.

SIGNS OF A PROMISING TARGET

- ✓ Altered in many patients (recurrent, not random)
- ✓ Concentrated in one subtype you can act on
- ✓ Sits at a hub in the protein network (STRING)
- ✓ Belongs to a pathway you can drug (Enrichr)
- ✓ Alteration links to worse outcomes

A TEXTBOOK SUCCESS: HER2

Observation → the ERBB2 (HER2) gene is amplified in a distinct subset of tumors.

Validation → blocking HER2 stops those cancer cells from growing.

Result → HER2-targeted antibodies transformed a once-aggressive subtype into a highly treatable one.

This is the same logic — observation to target — that your project practices.



PRECISION TREATMENT

Targets Already in the Clinic

Each subtype is treated differently precisely because each is driven by different molecules. These are real-world payoffs of the target-finding you're learning.



ER-Positive

Block the estrogen signal

- Endocrine therapy (e.g. tamoxifen, aromatase inhibitors)
- CDK4/6 inhibitors add cell-cycle blockade
- PIK3CA inhibitors for mutated tumors



HER2-Positive

Block the HER2 receptor

- Anti-HER2 antibodies (e.g. trastuzumab)
- Antibody-drug conjugates deliver chemo to HER2+ cells
- Often combined with chemotherapy



Triple-Negative

Exploit specific weaknesses

- Chemotherapy remains a backbone
- PARP inhibitors for BRCA-mutated tumors
- Immunotherapy for eligible patients



PUTTING IT TOGETHER

Your Workflow, End to End

1

Define

Write one clear, comparable research question across subtypes

2

Gather

Query your genes in cBioPortal; export to Excel / Sheets

3

Summarize

Tabulate frequencies; build clear charts of the patterns

4

Interpret

Run Enrichr & STRING; explain the biology you see

5

Conclude

State results, limitations, and treatment implications



A question template

*"Is gene **X** mutated or over-expressed more often in **triple-negative** than in **ER-positive** breast tumors — and what pathway does it belong to?"*

Specific, comparable, and answerable with the data you have. Make yours look like this.



THE FINISH LINE

Your Deliverable & Honest Science

YOUR POSTER SHOULD COVER

- **Research question** — The specific comparison you set out to make
- **Methods** — Databases, genes, and tools you used
- **Results** — Your tables, charts, and network findings
- **Limitations** — What your data can and cannot show
- **Implications** — What it might mean for treatment



Stay honest about limits

Correlation is not causation — a frequent mutation is a lead, not proof it drives cancer.

Databases have biases — cohorts may not represent all populations equally.

Sample sizes vary — small subgroups can produce unstable percentages.

Computation $\neq$ the clinic — a target still needs lab and trial validation.

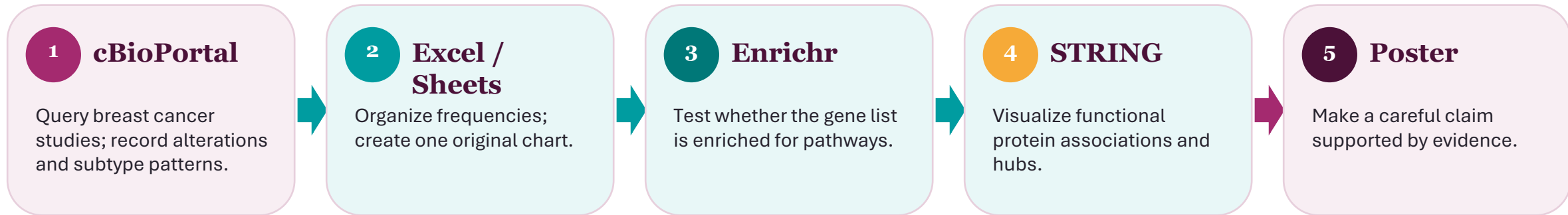
Ask a sharp question · Let the data answer · Report it honestly. That's it.



PROJECT WORKFLOW

The Project Workflow

Four linked steps turn database observations into a biological interpretation.



Core rule **Same workflow; different approved questions.**

This keeps the project manageable.



STRUCTURED CHOICE

Pick One Approved Project Track

Choose a question, but the dataset, workflow, and deliverables stay standardized.

1

Track A: Basal-like vs Luminal A

How do aggressive basal-like tumors differ from ER-driven luminal tumors?
Best beginner option; clean subtype comparison.

2

Track B: HER2-positive breast cancer

What molecular features characterize HER2-positive disease?
Clinically intuitive; clear ERBB2 amplification story.

3

Track C: TNBC target proposal

Which genes or pathways may represent targetable TNBC vulnerabilities?
Most research-oriented; requires careful interpretation.

1

STEP 1: CBIOPORTAL

What You Should Extract

Study

Breast Invasive Carcinoma, TCGA PanCancer Atlas or METABRIC

Cohort

Whole study first; then subtype group such as Basal-like or Luminal A

Genes

Use the approved panel; do not browse indefinitely

Alteration type

Mutation, amplification, deep deletion, structural variant when available

Frequency

Record percent altered for each gene

Evidence

Save one OncoPrint or relevant table screenshot

Export

Download or copy a small results table

Notes

Record date, dataset, filters, and subtype definition

Do not over-interpret

An altered gene is a lead. It is not proof that the gene caused the tumor or is a validated drug target.



Approved Gene Panels

A small panel prevents the project from becoming an uncontrolled database search.

Project track	Core genes	What they represent
Subtype comparison	ESR1, PGR, ERBB2, TP53, PIK3CA, GATA3, FOXA1, CCND1	receptors, drivers, luminal biology
HER2-positive	ERBB2, GRB7, TOP2A, PIK3CA, TP53, CCND1, EGFR	amplification, signaling, cell cycle
TNBC targets	TP53, BRCA1, BRCA2, PTEN, RB1, PARP1, CHEK1, AURKA, CDK1	DNA repair, checkpoints, proliferation
Endocrine resistance	ESR1, PIK3CA, FOXA1, GATA3, CCND1, CDK4, CDK6	ER signaling and cell-cycle escape

You may add 3–5 extra genes only after they can explain why the gene belongs in the analysis.

Make a Simple Analysis Table

The spreadsheet turns cBioPortal observations into a figure (table)

Gene	Track	Subtype A %	Subtype B %	Difference	Main alteration
TP53	Basal vs Luminal	High	Lower	Large	mutation
PIK3CA	Basal vs Luminal	Lower	Higher	Moderate	mutation
ERBB2	HER2+	High	Low	Large	amplification
BRCA1	TNBC	Variable	Lower	Contextual	mutation/deletion

Charts

- Bar chart comparing alteration frequencies across two groups
- Axis labels, study name, and subtype labels included
- Caption says what the pattern shows, not just what the figure is

Minimum analysis standard

You should compare at least five genes and produce one clean chart from its own table.

Build the Gene List for Interpretation

The gene list is the bridge from subtype comparison to pathway and network analysis.

Include genes that...

- show subtype-associated alteration patterns
- are recurrent across multiple tumors
- fit the biological question
- have interpretable gene symbols

Avoid lists that are...

- only 3–5 obvious markers
- hand-picked to force an answer
- full of genes students cannot explain
- mixed across unrelated questions

Target list size

- 20–50 genes is ideal
- 10 genes can work for STRING
- small marker panels are too weak for enrichment
- record exactly how the list was made

E

STEP 4A: ENRICHR

Use Enrichr for Pathway Enrichment

Enrichr asks whether genes in the list cluster in known biological processes or pathways.

1

Paste gene list

one gene symbol per line

2

Choose libraries

GO Biological Process, KEGG, Reactome

3

Record top terms

adjusted p-value, overlap genes

4

Interpret

what biology appears overrepresented?

What students record

- Top 3–5 enriched terms
- Adjusted p-value, not only the visual ranking
- Genes from the list that caused the hit
- A one-sentence biological interpretation
- Whether terms support the cBioPortal result

Interpretation example: “The TNBC-associated list is enriched for DNA repair and cell-cycle terms, suggesting a proliferation and genome-maintenance signal.”



STEP 4B: STRING

Use STRING for Protein Networks

STRING asks whether the proteins encoded by the genes are functionally connected.

1

Multiple proteins

paste the same gene list

2

Organism

Homo sapiens

3

Network

look for clusters, hubs, and isolated nodes

4

Enrichment

compare pathway terms with Enrichr

Appropriate claim

“These proteins form a connected network enriched for cell-cycle control, with TP53 and RB1 near central positions.”

Overclaim to avoid

“TP53 directly binds every protein connected to it.” STRING edges are evidence-supported associations, not always physical binding.

Final output

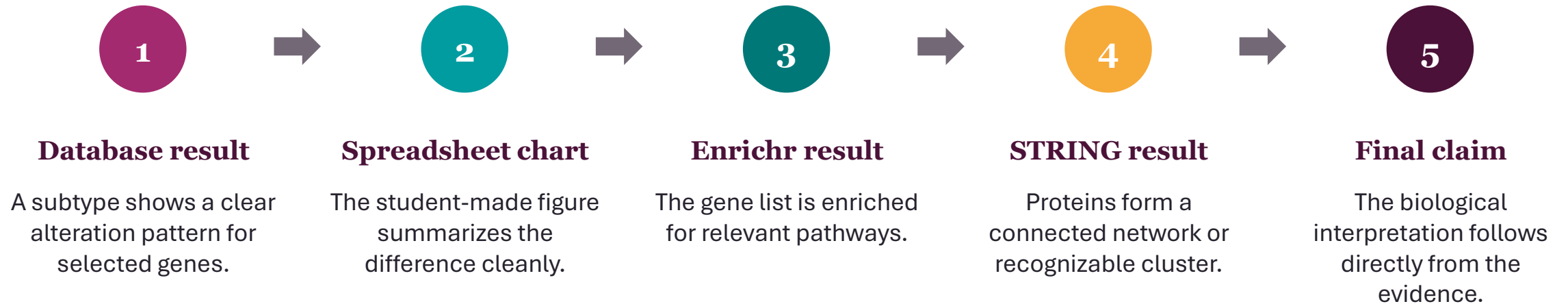
One network screenshot plus two sentences explaining the main cluster.



INTERPRETATION

How the Evidence Fits Together

A strong conclusion is supported by multiple kinds of evidence, not one.



Example integrated claim

“Basal-like tumors show more frequent TP53 alteration than Luminal A tumors, and the associated gene list is enriched for cell-cycle and DNA-damage pathways.”



SCIENTIFIC CLAIMS

Write Claims That Match the Evidence

The goal is not to sound certain. The goal is to be accurate.

Stronger language

- “is more frequently altered in...”
- “is consistent with...”
- “suggests involvement of...”
- “supports the hypothesis that...”
- “would require experimental validation”

Avoid overclaims

- “proves that...”
- “causes TNBC...”
- “will cure...”
- “is definitely a drug target...”
- “directly binds because STRING connects them”

An important sentence in the poster: “A limitation of this analysis is...”

Suggested One-Month Timeline

Each week should end with a concrete product, not just discussion.

1

Week 1

Biology + cBioPortal

Output: Research question
and chosen track

2

Week 2

cBioPortal data extraction

Output: Completed data table
and screenshots

3

Week 3

Sheets chart +
Enrichr/STRING

Output: Chart, pathway table,
network figure

4

Week 4

Interpretation + poster

Output: Final poster



SUCCESS CRITERIA

How to Know the Project Worked

Before presenting, you should be able to answer these questions.

- ✓ What exact breast cancer subtype comparison did you make?
- ✓ Which pathways were enriched in Enrichr?
- ✓ Which cBioPortal study and filters did you use?
- ✓ What did the STRING network add beyond the pathway table?
- ✓ Which genes were most different between groups?
- ✓ What is one limitation that prevents over-interpretation?
- ✓ What original chart did you make from your table?
- ✓ What experiment would you do next to validate the finding?

Final standard: one question, one database result, one chart, one enrichment result, one network result, and one honest conclusion.



TOOL REFERENCES

Where You Will Work

These are browser-based resources; no R, Python, or command line coding required.

cBioPortal

<https://www.cbioportal.org>

Cancer genomics study exploration, gene queries, OncoPrints, subtype filtering, group comparison.

Enrichr

<https://maayanlab.cloud/Enrichr/>

Gene-list enrichment analysis using GO Biological Process, KEGG, and Reactome libraries.

STRING

<https://string-db.org>

Functional protein association networks and enrichment analysis.

Google Sheets / Excel

[local or web-based spreadsheet](#)

Small tables, bar charts, frequency comparisons, and record-keeping.